

## **AI in Antibodies ( AI-Based Nanobody Structure Prediction and Its Downstream Application in Antigen Docking and Epitope Recovery )**

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Internship Duration: From: 15/7/2026

To: 31/8/2026

### **Job Responsibilities:**

1. Review nanobody/VHH structure, CDR regions, and AI-based antibody structure prediction methods.
2. Benchmark SimpleFold, IgFold, and NanoBodyBuilder2 using a common VHH structural dataset.
3. Perform antigen–nanobody docking and evaluate docking quality using HDOCK and DockQ.
4. Analyze epitope recovery and contribute to interpretation of results and research manuscript preparation.

### **Expected Outcomes:**

1. Compare framework and CDR-H3 structural accuracy across the three prediction models.
2. Determine whether differences in nanobody structural accuracy affect downstream docking performance.
3. Assess recovery of experimentally observed antigen epitopes from predicted docking poses.
4. Develop practical skills in computational structural biology, scientific analysis, and research-paper preparation.

***Menna Nabil***

Student Signature

Supervisor Signature

Program/Center Director Signature

Nanobodies are compact single-domain antigen-binding fragments derived from camelid heavy-chain-only antibodies. Their small size, high stability, and ability to recognize recessed or sterically restricted epitopes make them valuable tools in therapeutics, diagnostics, structural biology, and epitope-focused research. Because their binding behavior depends strongly on the conformation of the CDR loops, especially CDR-H3, accurate structure prediction is important for downstream applications such as docking and epitope mapping. Among the structural regions of a nanobody, CDR-H3 is particularly important because it often makes substantial contributions to antigen recognition while also exhibiting high variability in sequence, length, and three-dimensional conformation. This combination makes CDR-H3 one of the most difficult regions to predict accurately. Consequently, a structure-prediction model may reconstruct the conserved VHH framework correctly while still producing a binding loop that differs considerably from the experimentally observed conformation. Such errors may become especially important when the predicted structure is later used for applications such as antigen docking or identification of the antigen epitope.

Three structure-prediction models were evaluated: SimpleFold (a general-purpose protein folding model), IgFold (an antibody-specific model built on a large antibody language model), and NanoBodyBuilder2 (a model developed specifically for nanobody structures). Their comparative accuracy was then carried forward into a downstream docking and epitope-recovery pilot study to test whether benchmark-level differences translate into functionally meaningful outcomes.



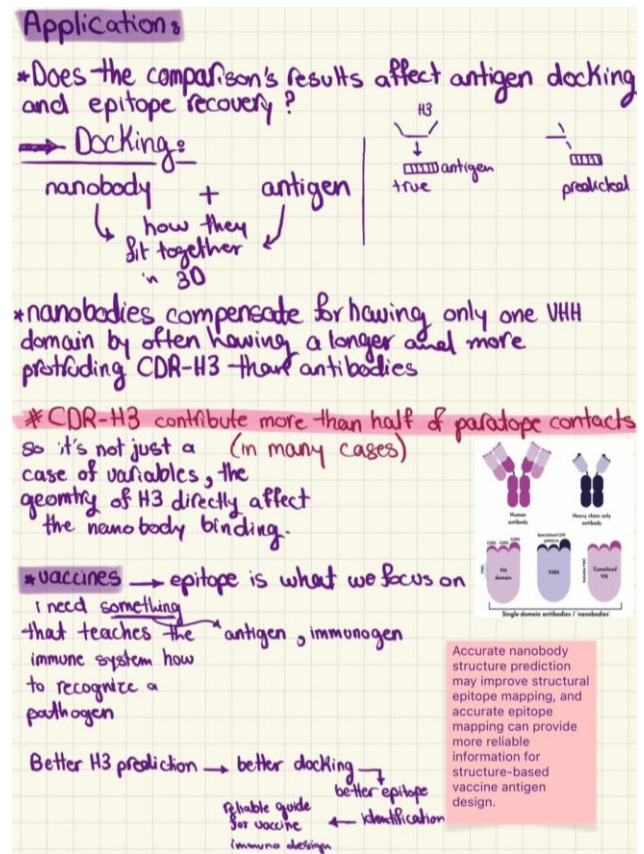
## Assigned Tasks:

- Review nanobody structure, CDR regions, and CDR-H3 significance.
- Review AI-based antibody and nanobody structure prediction methods.
- Prepare and validate the VHH benchmark dataset.
- Benchmark SimpleFold, IgFold, and NanoBodyBuilder2.
- Compare framework and CDR-H3 structural accuracy.
- Investigate SimpleFold conformational sampling and uncertainty.
- Perform antigen–nanobody docking.
- Evaluate docking poses using DockQ.
- Evaluate antigen epitope recovery.
- Interpret the relevance of structural accuracy to downstream biomedical applications.
- Contribute to preparation of the research manuscript.

## Work Accomplished and Outcomes:

We started with a review of antibody versus nanobody structure, VHH architecture, the complementarity-determining regions (CDRs) and the particular structural significance of CDR-H3, antibody-specific versus general-purpose protein structure predictors, molecular docking, and epitope mapping. This review grounded the subsequent benchmarking and application work in the relevant structural-biology and computational literature.

A common set of 597 experimentally determined VHH structures was used to evaluate SimpleFold, IgFold, and NanoBodyBuilder2. Structural accuracy was assessed using framework and CDR-specific RMSD, with particular emphasis on CDR-H3.



As SimpleFold is a general-purpose, generative structure predictor rather than an antibody-specific tool, additional analysis was performed on its behavior, including generative sampling, repeated conformations, uncertainty and sample spread, sampling prior coverage, and an ESM2 representation probe. Implementation detail beyond this summary is reserved for the accompanying research manuscript

| Model            | Median CDR-H3 RMSD (Å) | Framework RMSD (Å) |
|------------------|------------------------|--------------------|
| SimpleFold       | 2.944                  | 0.821              |
| IgFold           | 2.885                  | 0.827              |
| NanoBodyBuilder2 | <b>2.484</b>           | <b>0.793</b>       |

The project therefore addressed two connected questions. First, can a general-purpose protein structure predictor such as SimpleFold perform competitively with models developed specifically for antibodies or nanobodies? Second, do differences observed at the structural level, particularly in CDR-H3 accuracy, affect the usefulness of the predicted structures in downstream applications? The study was designed to move beyond RMSD-based benchmarking by linking structural prediction accuracy to antigen docking and epitope recovery.

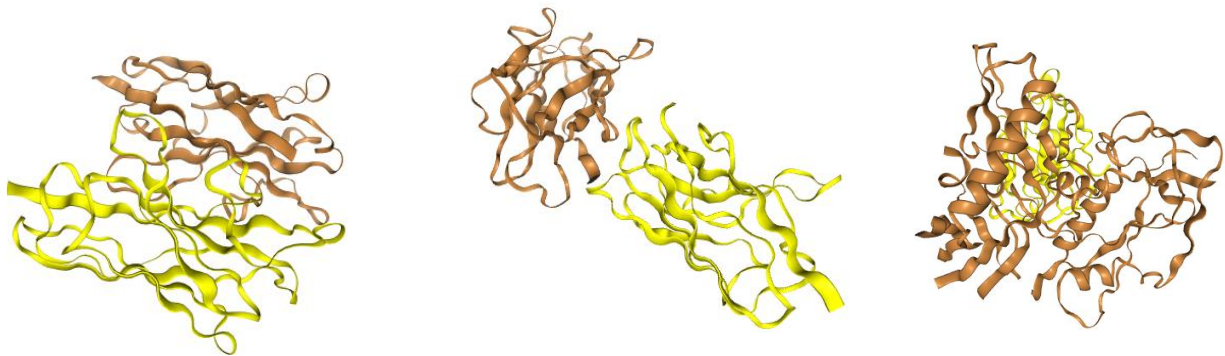
A pilot study was designed to test whether structural prediction differences observed in the benchmark actually matter biologically. The case selected was the 7R73 nanobody bound to HIV-1 gp120 (chain A). Four nanobody inputs were compared: the experimental VHH structure (positive control), and the structures independently predicted by SimpleFold, IgFold, and NanoBodyBuilder2. All four were docked to the same gp120 antigen using template-free HDOCK, keeping the antigen, target case, and docking protocol fixed so that the input nanobody structure was the only variable under test.

| VHH Input        | Top-1 DockQ | LRMSD (Å) | f <sub>nat</sub> |
|------------------|-------------|-----------|------------------|
| Experimental     | 0.986       | 0.640     | 1.000            |
| NanoBodyBuilder2 | 0.740       | 2.926     | 0.700            |
| IgFold           | 0.022       | 35.767    | 0.000            |
| SimpleFold       | 0.009       | 60.919    | 0.000            |

*NanoBodyBuilder2 produced the only predicted VHH structure that enabled recovery of a near-native Top-1 antigen-binding pose.*

The direct three-model comparison was performed on the 588 targets successfully predicted by all three methods. Median CDR-H3 RMSD was 2.944 Å for SimpleFold, 2.885 Å for IgFold, and 2.484 Å for NanoBodyBuilder2, while median framework RMSD was 0.821 Å, 0.827 Å, and 0.793 Å, respectively. These results showed that all three models were highly competitive in reconstructing the conserved VHH framework, whereas the main advantage of the specialized models appeared in the more variable CDR-H3 region. NanoBodyBuilder2 achieved the strongest overall H3 accuracy, while SimpleFold remained surprisingly competitive despite not being specifically designed for antibody or nanobody prediction.

This setup was deliberately designed as a controlled re-docking experiment. The gp120 antigen was retained in its experimentally observed bound conformation, while only the VHH structure was changed between experiments. The experimental VHH was included as a positive control to confirm that the docking procedure could recover the known interaction when supplied with the native-like nanobody conformation. HDOCK was run in template-free mode without binding-site restraints, preventing the experimentally known interface from being supplied directly to the docking algorithm. The Top-10 docking poses were evaluated against the known experimental complex using DockQ. Top-1 DockQ measured ranking performance, while the best DockQ among the Top-10 assessed whether a near-native pose was sampled.



The experimental VHH produced a Top-1 DockQ score of 0.986, confirming that the docking pipeline could recover the native binding mode under ideal structural input. Among the predicted structures, NanoBodyBuilder2 achieved a Top-1 DockQ score of 0.740 with a ligand RMSD of 2.926 Å and recovered 70% of native contacts. In contrast, the Top-1 IgFold and SimpleFold poses obtained DockQ scores of 0.022 and 0.009, respectively. For this pilot complex, NanoBodyBuilder2 was therefore the only predicted structure that enabled recovery of a near-native Top-1 binding pose.

The experimental antigen epitope was defined using a 5 Å heavy-atom contact cutoff. Predicted docking poses were compared with this epitope using recall, precision, and Jaccard similarity.

| Input            | Correct Epitope Residues | Recall | Jaccard |
|------------------|--------------------------|--------|---------|
| Experimental     | 26 / 26                  | 1.000  | 0.963   |
| NanoBodyBuilder2 | 20 / 26                  | 0.769  | 0.645   |
| IgFold           | 2 / 26                   | 0.077  | 0.042   |
| SimpleFold       | 0 / 26                   | 0.000  | 0.000   |

The experimentally observed gp120 epitope contained 26 residues under this contact definition. Re-docking the experimental VHH recovered all 26 residues, whereas NanoBodyBuilder2 recovered 20 of 26, corresponding to a recall of 0.769 and a Jaccard similarity of 0.645. The Top-1 IgFold pose recovered only 2 of the 26 experimental residues, while the Top-1 SimpleFold pose recovered none. Thus, for the 7R73 pilot, the ordering observed in structural accuracy was also reflected in docking quality and recovery of the experimentally observed antigen epitope. This result also illustrates the potential relevance of accurate nanobody prediction to structure-guided biomedical research. Reliable identification of an antigen epitope can help characterize which antigenic regions are recognized by a binder and may therefore provide structural information useful for deciding which regions should be preserved, exposed, or stabilized during antigen and immunogen design. The present result represents a single-complex pilot and therefore requires validation across additional nanobody–antigen systems before this relationship can be generalized.

## Learning Outcomes and Gained Skills:

### 1- Technical Skills:

Throughout the internship, I developed practical technical skills in protein structure analysis and nanobody/VHH structural biology, with particular emphasis on CDR-H3. I gained experience in RMSD calculation and structural alignment, working with PDB structures, and performing Python-based structural data analysis. I also learned how to conduct molecular docking using HDock, evaluate docking quality using DockQ, analyze antigen–nanobody contacts and epitope recovery, and apply AI-based protein structure prediction tools to comparative structural studies.

## **2- Research Skills:**

The internship also strengthened my research skills through extensive literature review, identification and formulation of research gaps, and the design of controlled computational experiments. I learned how to distinguish between experimental controls, conformational sampling, and docking ranking, and how to interpret computational results carefully without overgeneralizing conclusions. In addition, I developed skills in scientific visualization, IEEE-style academic writing, and collaborative preparation of a research manuscript.

## **Conclusion:**

The internship provided valuable practical experience in applying artificial intelligence and computational structural biology to nanobody research. The study demonstrated that conventional structural evaluation alone does not fully capture the potential downstream utility of a predicted nanobody structure. In the 7R73 pilot, differences in CDR-H3 structural accuracy were reflected in both antigen-docking quality and epitope recovery, highlighting the importance of evaluating structure-prediction models not only by RMSD-based metrics but also by their performance in downstream structural applications.

Future work will extend the current pilot analysis to a larger and more diverse set of experimentally resolved nanobody–antigen complexes in order to determine whether the observed relationship between CDR-H3 structural accuracy, DockQ performance, and epitope recovery is reproducible across different targets and antigen classes. This broader evaluation will make it possible to test whether lower CDR-H3 RMSD is consistently associated with higher docking quality and more accurate recovery of experimentally observed antigen-contact residues, rather than being specific to the 7R73 case. Correlation analyses across multiple complexes will also help quantify the strength of this relationship and identify cases in which good structural accuracy does not necessarily translate into good downstream docking performance.

Further work will also investigate the effect of conformational sampling, particularly for SimpleFold, which can generate multiple structural predictions for the same VHH sequence. Evaluating several conformers per target may reveal whether alternative sampled CDR-H3 structures can improve docking and epitope recovery, and whether internal sampling variability can be used as a practical indicator of prediction uncertainty. In addition, the robustness of the conclusions will be tested using alternative docking platforms and protocols to determine whether the observed trends depend on HDOCK specifically or remain consistent across different docking algorithms.

Finally, the downstream implications of reliable epitope recovery will be explored in the context of structure-guided antigen and immunogen design. If accurate nanobody structures consistently lead to more reliable identification of antigenic contact regions, predicted nanobody–antigen interactions could provide useful structural information for identifying epitopes that may be preserved, stabilized, or preferentially exposed in candidate immunogens. Such work would extend the project from benchmarking structure-prediction models toward evaluating their practical value in structure-guided biomedical and vaccine research.